Real experimental conditions of Rydberg gate

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Abstract

In this note, I intend to investigate the experimental conditions of Rydberg gate based on different gate schemes

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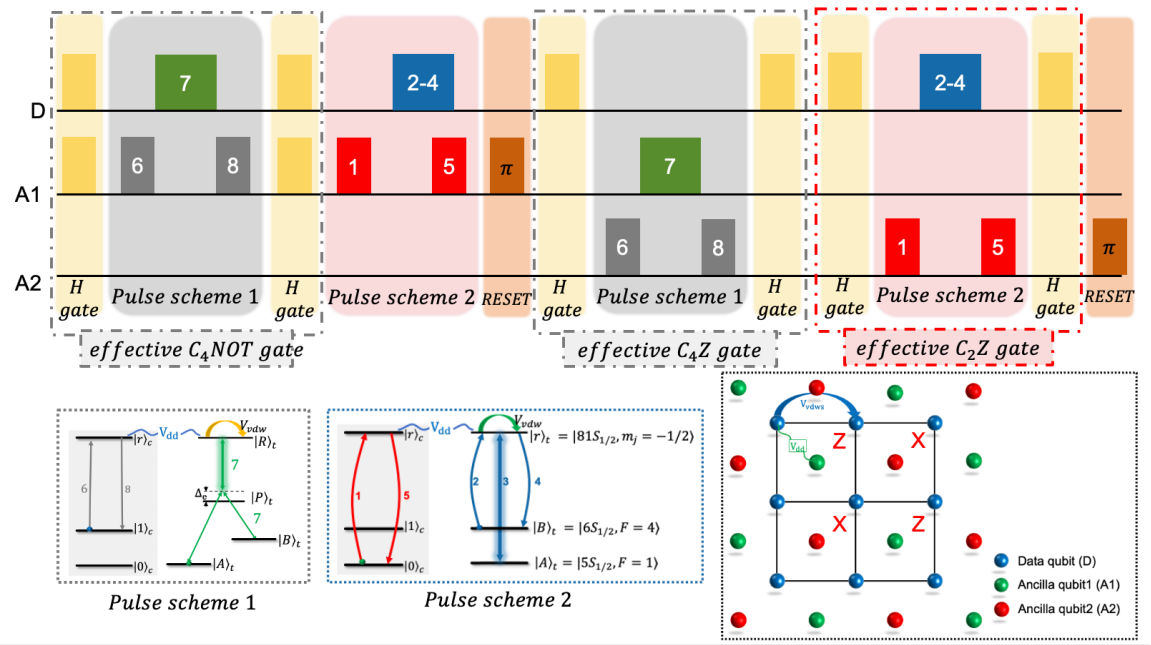
AMS subject classification. 65M70, 65M12

1 Species of Rydberg atoms

(i) In this work, we intend to realize a measurement-free error correction scheme. The model and pulse sequence are shown below:



In this model, we determine to use three different atomic species. The data qubit is 87-Rb atom, the ancilla qubit 2 is 133-Cs atom and the ancilla qubit 1 is 85-Rb atom.

2 Interactions between atoms

In this work, we assume 3 different atomic species, 87-Rb, 133-Cs and 85-Rb. We will describe the interactions in heternuclear and homonuclear cases.

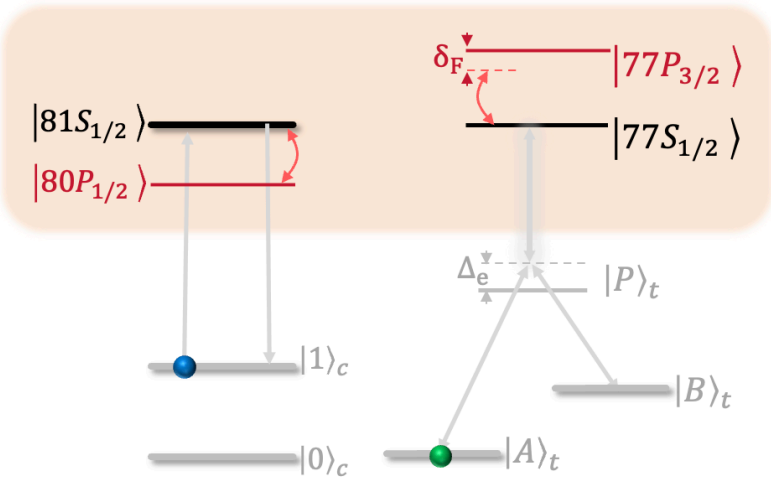
When discussing the distance between atoms, we should also pay attention to the Le Roy radius(R_{LR}). When the distance between atoms is smaller than the Le Roy radius, there will be electron cloud overlap, which triggers the exchange interactions, charge transfer, or inelastic

collisions. These processes cause decoherence, spectral broadening, or unwanted molecular formation, disrupting quantum operations. Thus, we need to make sure that the distance between atoms is larger than the Le Roy radius to avoid unwanted and uncontrolled short-range effects.

2.1 Heteronuclear interactions between 87-Rb and 133-Cs

We consider the 133-Cs atom excited to the Rydberg state $|r\rangle_c = |81S_{1/2}, m_j = -1/2\rangle$ as control qubit and the 87-Rb atom excited to the Rydberg state $|R\rangle_t = |77S_{1/2}, m_j = 1/2\rangle$ as target qubit. Due to the interaction between the two atoms the Rydberg states in both atoms will be coupled to other neighboring state(s) and the most dominant interaction channel and corresponding figure are shown below, with energy defect $\delta_F = 2\pi \times 2$MHz, dipole-dipole interaction coefficient $C_3 = 2\pi \times 10$ GHz.

$$|81S_{1/2}, m_j = -1/2; 77S_{1/2}, m_j = 1/2\rangle \rightarrow |80P_{1/2}, m_j = 1/2; 77P_{3/2}, m_j = 3/2\rangle$$



The Le Roy radius for the heteronuclear interaction is $R_{\mathrm{LR}} = 2$ um and the distance is required to be larger than the Le Roy radius, i.e., $d > R_{\mathrm{LR}} = 2\mu m$. The possible leckage channel is shown below:

α	ω_1	^{87}Rb				ω_2	^{133}Cs				$C_3/2\pi$	$\delta_F/2\pi$	χ	$C_6/2\pi$
		n_1	ℓ_1	j_1	m_{j1}		n_2	ℓ_2	j_2	m_{j2}				
1	σ^+	76	1	1.5	1.5	σ^+	81	1	0.5	0.5	10.9	1.87	4.650×10^{-2}	5.812
2	π	77	1	0.5	0.5	σ^+	80	1	0.5	0.5	5.88	0.214	2.201×10^{-1}	27.52
3	σ^+	77	1	1.5	1.5	σ^+	80	1	0.5	0.5	10.0	0.002	$3.656 \times 10^{+1}$	4570
4	π	78	1	0.5	0.5	σ^+	79	1	0.5	0.5	0.105	-0.437	1.927×10^{-3}	-0.241
5	σ^+	78	1	1.5	1.5	σ^+	79	1	0.5	0.5	0.190	-0.64	2.374×10^{-3}	-0.297

2.2 Homonuclear interactions between 87-Rb atoms

The homonuclear interaction between two 87-Rb atoms excited to the Rydberg state $|R\rangle_t = |77S_{1/2}, m_j = 1/2\rangle$ amost have no leckage channel. The van der Waals coefficient is $C_6 = 2\pi \times 2036$ GHz μm^6.

α	ω_1	^{87}Rb				ω_2	^{87}Rb				$C_3/2\pi$	$\delta_F/2\pi$	χ	$C_6/2\pi$
		n_1	ℓ_1	j_1	m_{j1}		n_2	ℓ_2	j_2	m_{j2}				
1	σ^+	76	1	1.5	1.5	σ^+	76	1	1.5	1.5	11.2	16.84	6.657×10^{-4}	67.26
2	π	77	1	0.5	0.5	π	77	1	0.5	0.5	4.29	-15.40	2.784×10^{-4}	-96.78
3	σ^+	77	1	1.5	1.5	σ^+	77	1	1.5	1.5	12.5	-15.82	7.873×10^{-4}	-88.39
4	π	78	1	0.5	0.5	π	78	1	0.5	0.5	0.060	-46.79	1.294×10^{-6}	-0.006
5	σ^+	78	1	1.5	1.5	σ^+	78	1	1.5	1.5	0.197	-47.20	4.178×10^{-6}	-0.007

3 X-error correction

We assume that there is a single X error on the center data qubit. The error syndrome will be mapped to the ancilla qubit 1 through a effective C_4NOT-gate with four data qubits in one plaquette are perform as control qubits and the ancilla qubit 1 in the center of the plaquette is the target qubit. The realization of the effective C_4NOT-gate is shown in the first dashed black box in the figure shown above.

Thus, we can concentrate on realizing a CNOT_4 gate with one control qubit controlling four data qubits. The gate scheme 1 describes the gate model of the CNOT_4 gate:

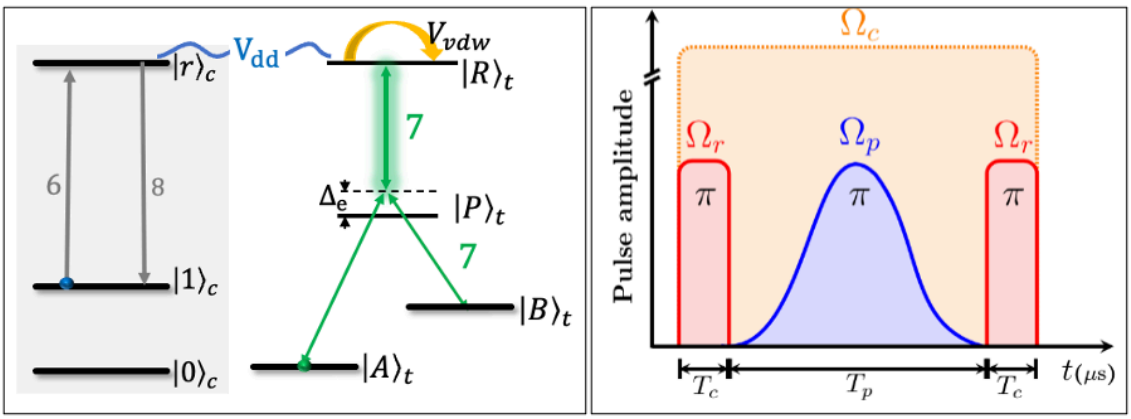
(i) In the energy level diagram:

(1) The energy levels of control atom (Cs) are $|0\rangle_c = |6S_{1/2}, F = 4\rangle$, $|1\rangle_c = |6S_{1/2}, F = 3\rangle$, $|r\rangle_c = |81S_{1/2}, m_j = -1/2\rangle$(with lifetime $\tau_c = 548\mu s$).

(2) The energy level of target atoms(87-Rb) are $|R\rangle_t = |77S_{1/2}, m_j = 1/2\rangle$(with lifetime $\tau_t = 505\mu s$), $|P\rangle_t = |6P_{3/2}\rangle$(with lifetime $\tau_P = 0.131\mu s$), $|A\rangle_t = |5S_{1/2}, F = 1\rangle$, and $|B\rangle_t = |5S_{1/2}, F = 2\rangle$.

(ii) In the coupling process: $\Omega_p = \Omega_{\max} \sin^2\left(\frac{\pi}{T_p}(t - T_c)\right)$, $\Omega_{\max} = \Omega_r = 2\pi \times 50$ MHz; $\Omega_c = 2.5\Omega_{\max} = 2\pi \times 125$MHz; and the detuning $\Delta = 2\pi \times 1200$ MHz

(iii) The interaction strength: the interaction strength between atoms is denoted by V_{dd} and V_{vdws}. We assume that the distance between control and target atoms is $d = 6$ um and the distance between target qubits is $6\sqrt{2}$.



3.1 Model

Bibliography